

Episode 18: Once Upon a Time... (13) *Jem* and "Brown Carbon"

I took 10 days vacation. Not travelling anywhere, just living normal daily life in Tokyo, cleaning up the house and surroundings, repairing broken appliances.

Yesterday was disposing of videotapes from home. VHS tapes and - 8mm video!

When 8mm video first appeared, the picture quality was decent, and above all I was charmed by the small size of 8mm video tapes that took up so little space, so I suddenly became an "8mm video enthusiast" and recorded like crazy. So there are tremendous amounts of videotapes at both home and my parents' house. But eventually 8mm video became a distant past thing too... Ah, it followed the same fate as beta, Sony (bitter laugh).

So I disposed of the home batch together, but most of it was movies, dramas, and anime broadcast on TV. Now with movies and anime, when you want to watch them, TSUTAYA and other DVD rentals are well-stocked, so there's no need to keep old stuff. ...Or rather, the 8mm video VCR itself at home is already at the end of its life (cries).

Still, I excavated various treasures. Special programmes from "Denpa Shonen," the late-night broadcast Whychoy Production drama "Johin Driver" (laughs). These will also be gradually dubbed to HDD-DVD recorder (laughs).

Well then.

After the domestic theatrical works *GeGeGe no Kitaro: Yokai Daisenso* and *Hokuto no Ken: Seikimatsu Kyuseishu Densetsu* continued, my 1986 work was still mainly collaborative work.

That year I handled *Transformers*, *Muppet Babies*, *Glo Friends* (per-episode colour direction and inspection for all), numerous collaborative work commercials, pilot films, and *Jem*.

This work features a BAND of four girls named "Jem," "Shana," "Kimber," and "Aja" (though they're about 20?) as protagonists. They tour worldwide performing, getting into predicaments through various interference from rival band led by "Pizzazz," but somehow get through it every time (laughs).

I was assigned to handle this work. Well, ultimately, from Episode 4 onward, all colour-related work would be handed over completely to subcontracting production companies, but those initial 3 episodes were tough again. Being a girls' band story, naturally there were 2-3 costume changes per episode. Naturally around the eyes were sharp eyeliner and eyeshadow. All naturally colour trace. And what was most difficult - this work was a "brown carbon" work.

Back then, we used "trace machines" to transfer in-between animation to cels. The mechanism was almost the same as "Print Gocco." Line drawings (in-between animation) drawn on paper were heat-transferred to cels from the back side using special carbon paper. This carbon was actually troublesome.

Regular anime works were produced using "black" carbon. But depending on work content and nature, sometimes non-black coloured carbon paper was used. Mechanically, different carbon colours meant different transferred main line colours. There were several so-called "colour carbons" - "red," "white," "blue," "green," etc., and one of those colours was "brown." For soft-toned works or to make female character outline main lines look softer, this "brown carbon" was very effective. So it was perfect for *Jem*. But this "brown carbon" had a big problem.

When transferring with trace machines, only main lines transfer, but naturally untransferred areas are also heated while covered by carbon paper. As a result, oil from the carbon paper faintly sticks to areas without lines. This was a major problem.

Water-based cel paints are water-based, so literally water and oil. The paint would be beautifully repelled.

This was what I heard from sales staff at Josai Duplo, which handled trace machines and dedicated carbon paper exclusively back then - compared to regular "black carbon," "colour carbons" had lighter colours, making them technically difficult to manufacture. They couldn't achieve the clean sharpness of "black carbon" when transferred, plus oil adhesion increased.

Incidentally, due to that "manufacturing technical difficulty," unit costs were several yen higher than black carbon, about 25-26 yen per sheet I think.

Despite higher unit costs, there were many product precision inconsistencies - when good, oil adhesion was almost as little as "black carbon," but that was actually rare. When bad, it was thoroughly terrible (explosion). And this *Jem* launch period was directly hit by that "bad period" (bitter laugh).

So finishing work was difficult. "Brown carbon" itself had already been used for *Muppet Babies* and *Glo Friends*, but those had rounded characters without much shadowing, so we could somehow manage. But this *Jem* was a shadowed work with lots of detailed colour divisions - facial makeup, costumes, instruments, etc. Plus colour trace. With solid line colour divisions, you could somehow place paint hooking onto those carbon lines, but with colour trace, painted surfaces had no grip - just pure paint repelling.

Furthermore, compared to "black carbon," in-between animation didn't transfer cleanly, so transferred lines were thin or broken, often requiring careful trace correction before painting. Therefore, to transfer more solidly, trace machine heating time was set slightly longer, but if slightly too strong, "burning" would occur in areas without lines, creating faint carbon stains. Painting without noticing these stains would paint over the dirt, resulting in NG. With moderate dirt and darker coloured paints, it

wasn't very noticeable and helped, but this *Jem* had lots of bright, light colours, causing unplanned brown gradations on hair and costumes, leading to retakes.

The countermeasure was wiping each piece with alcohol or some weak chemical before painting (bitter laugh). Hmm.

Speaking of which, in the theatrical *Transformers* I mentioned earlier, there was a female robot character, and only this character was processed with "brown carbon" while regular robots (using "black carbon") had different trace machine settings for each carbon type in the same in-between animation, so they couldn't be transferred at once - quite difficult.

As a further aside, later when I did *Sailor Moon S Movie*, the appearing Snow Queen and snow fairy (?) used "blue" and "white" carbons. But these colours had transparent main lines that were overpowered by painted paints, making lines invisible. So we actually layered another cel with just colour carbon traced lines over the actually painted cel for photography.

Due to such circumstances, changing main line colours was quite risky back then. With current digital painting, changing main line colours to create works is normal (laughs). This *Jem* would probably become something interesting if remade with today's technology.

Now, amid such "collaborative work" days, I was assigned a new work. A domestic work - and a TV series! Saturday night 7 PM golden time broadcast! The character sheets and storyboard cover handed to me had the large title "*Saint Seiya*" written on it.

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